

Detecting PTSD from one minute of six-channel dry EEG: a confounder-aware analysis of a multi-batch dataset

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Abstract

Background. Resting-state EEG has been proposed as an objective aid to the diagnosis of post-traumatic stress disorder (PTSD), and low-cost dry headsets would make such a test widely available. Small clinical datasets collected in several batches, however, invite models that learn how the data were recorded rather than who was recorded.

Methods. We analysed one minute of eyes-closed resting EEG from a six-channel dry headset in 229 people: 25 patients with combat-related PTSD, 130 healthy controls from three recording batches including 23 adults over 65, and 74 patients with somatoform disorders. We audited the files before modelling, built a logistic regression on 23 gain-independent spectral features, and evaluated it with nested leave-one-group-out validation in which all choices were made inside the training folds. Every analysis protocol was written down before it was run. We then tested the result against export format, recording quality, age, behaviour, unseen recording batches and label permutation.

Results. The EDF export format almost coincided with the diagnosis, and a rule based on the format alone reached an AUC of 0.985. Our model separated PTSD from controls under 65 with an out-of-sample AUC of 0.86 and detected 84% of patients at the 0.5 threshold. It separated patients from older adults reliably, mainly through the age-related decline of alpha. Within a single export format the model still worked, and shuffling labels inside that format never reproduced the result. Separation from young controls, however, relied on the temporal leads next to the reference electrode, and dropped when a recording batch was unseen. The features shifted in PTSD were shifted in the same way in somatoform patients.

Conclusions. One minute of dry EEG carries information that separates this clinical cohort from healthy controls, but the most robust part of it is not specific to PTSD and part of it depends on recording conditions. We report what the model measures and what it does not, and list what a dataset would need for a trauma-specific marker.

Keywords: PTSD; resting-state EEG; dry electrodes; confounders; shortcut learning; nested cross-validation; mismatch negativity

1. Introduction

PTSD is diagnosed by clinical interview and questionnaires. An objective physiological measure could support screening and the monitoring of treatment, and resting-state EEG is an attractive candidate because it is cheap and quick to record. In veterans with PTSD, EEG recorded during cognitive tasks showed faster and weaker alpha and a flatter power spectrum than in controls (Kovacevic et al., 2025), and a meta-analysis of event-related potentials found changes in components such as P3 (Karl et al., 2006). These effects are modest and were mostly obtained with dense gel electrode systems in laboratory conditions.

Consumer-grade dry headsets with a handful of electrodes change the picture. The signal is noisier, the montage is sparse, and data are often collected in batches with different devices, software versions or export settings. When such a batch happens to coincide with a diagnostic group, a classifier can reach high accuracy by learning the batch. This failure, known as shortcut learning (Geirhos et al., 2020), is

hard to see in a single accuracy number, and small samples make the accuracy itself uncertain (Varoquaux, 2018).

This paper describes our solution to a hackathon task: estimate the probability of PTSD from a six-channel dry EEG recording. Our aims were to find out what the data can support, to build a model whose behaviour can be explained, and to test it against every confounder we could measure. We treat the audit as the main result and the classifier as a tool to probe the data.

2. Data

Recordings were made with a NeuroPlay-6C headset with dry electrodes at O1, T3, Fp1, Fp2, T4 and O2, referenced to the left earlobe, at about 125 Hz. Each session contained one minute of rest with eyes closed and, for most subjects, five Schulte tables solved with eyes open. A 2 Hz metronome with every eighth beat deviant played during the recordings, without stimulus markers. Table 1 lists the groups.

Table 1. Groups in the development data.

Group	Subjects	Recordings	Age
PTSD (combat-related)	25	rest and five Schulte tables	unknown, 18–45 by design; men
Controls, original release	67	rest and five Schulte tables	19–58
Controls, supplementary batch	40	rest only	17–41
Controls aged 65+	23	rest and the first table	65–70
Somatoform disorders	74	rest and five Schulte tables	unknown, 18–45 by design; both sexes

Some recordings were byte-identical copies stored under different subject IDs. Merging subjects that shared any recording gave 221 independent groups, and every data split in this work respects these groups. The age of the patient groups and the sex of all subjects other than the PTSD group were not available.

3. Methods

3.1 Data audit

Before modelling, we inspected file headers, amplitude resolution, sampling rate, recording length, channel correlations and spectra of every file. The aim was to find any property of the files that could predict the label without reflecting brain activity.

3.2 Preprocessing and features

Signals were resampled to 125 Hz and low-pass filtered at 40 Hz, because above 40 Hz the spectrum depended on the export format. Zero padding at the end of some files was trimmed. Each recording was cut into 4-second windows with a 2-second step, and windows with a flat signal, clipping, very large amplitude, outlying variance or a copied channel were rejected per channel with the same thresholds for everyone. The power spectrum of each channel was averaged in the logarithmic domain over the retained windows. O2 was excluded because in one batch it behaved like a frontal channel.

From the rest recording we computed 23 features on O1, Fp1, Fp2, T3 and T4: the relative power of theta (4–8 Hz), alpha (8–13 Hz) and beta (13–30 Hz), the frequency and height of the alpha peak at O1 above the aperiodic background, the slope of the aperiodic spectrum (Donoghue et al., 2020), and frontal alpha asymmetry. All features are relative and do not depend on the overall gain of a channel. Recording quality was deliberately not used as a feature, because patients lost more windows to artefacts than controls.

Let $S(f)$ be the spectrum of a channel. The relative power of a band b is the logarithm of its share in the 4–30 Hz range:

$$r_b = \log_{10} (P_b / P_{4-30}), \quad P_b = \sum_{f \in b} S(f) \quad (1)$$

The aperiodic background is a straight line in logarithmic coordinates, fitted by least squares on 3–30 Hz with the alpha range 7–14 Hz left out. Its slope χ is the spectral slope feature, and the alpha peak is the largest excess of the spectrum over this line between 7 and 13 Hz:

$$\log_{10} S(f) \approx \beta - \chi \log_{10} f \quad (2)$$

Frontal alpha asymmetry compares alpha power at the two frontal poles:

$$\text{FAA} = \ln P_\alpha(\text{Fp2}) - \ln P_\alpha(\text{Fp1}) \quad (3)$$

3.3 Model and validation

The classifier is an L2-regularised logistic regression with balanced class weights, preceded by median imputation and standardisation. With 25 patients, a simple linear model is easier to interpret and less likely to overfit. Its output is mapped to a probability by Platt scaling (Platt, 1999).

Performance was estimated with nested cross-validation (Cawley and Talbot, 2010). The outer loop leaves out one independent group at a time. Inside each training fold, a group-stratified five-fold cross-validation chooses the feature set, the strength of regularisation, whether somatoform patients are used as negatives, and the weight given to older adults and somatoform patients. The selection criterion mirrors the official scoring formula. It combines the AUC of PTSD against controls under 65 with the AUC against the specificity groups (older adults and somatoform patients), both computed on inner out-of-fold predictions:

$$J = 30 \cdot \max(0, 2AUC_{\text{controls}} - 1) + 20 \cdot \max(0, 2AUC_{\text{specificity}} - 1) \quad (4)$$

The selected model gives a score $\hat{p}$. Platt scaling with balanced class weights maps it to a probability, and its offset is then moved by τ so that, at $P = 0.5$, sensitivity on PTSD equals specificity on the specificity groups in the inner predictions:

$$P = \sigma(a \cdot \text{logit } \hat{p} + b - \tau), \quad Se(\tau) = Sp(\tau) \quad (5)$$

Outer predictions were never used for any choice. Confidence intervals come from 2000 bootstrap resamples of groups. All settings are listed in Appendix A.

3.4 Pre-registration

Each change to the method was described as a numbered protocol with its decision rule before it was run, and the results of all protocols are kept, including those that failed. Eight protocols were run in total; the submitted model is protocol 8.

3.5 Confounder and robustness analyses

We trained diagnostic models in the same folds on export metadata alone and on recording quality alone. We evaluated the model within a single export format, replaced the temporal features with training medians at test time, and retrained without the temporal leads. We tested whether the features predict age within the controls and whether the probability follows age or Schulte solve time. Two further checks were fixed in advance. In the first, each control group was removed from the whole procedure and then predicted as new data. In the second, labels were shuffled only among subjects of one export format and the whole nested procedure was repeated 60 times.

3.6 Auditory response without stimulus markers

Because the metronome period is fixed, each recording can be folded into 4-second cycles of eight beats. Responses to every beat appear at multiples of 2 Hz in the spectrum of the average cycle, and a response to the deviant appears at the remaining multiples of 0.25 Hz. We searched for the timing on even cycles and tested it on odd cycles, and a random-shift control repeated the whole search 2000 times. The method was checked on the open ERP CORE mismatch negativity data (Kappenman et al., 2021), filtered to imitate the headset, and in a hybrid simulation in which a response of known size was added to real recordings after removing their own response with the plus-minus method (Schimmel, 1967). The deviant test was run once, as a confirmatory test with a pre-set threshold.

4. Results

4.1 The export format follows the diagnosis

Each subject had been exported in one of three EDF formats, which differ in amplitude resolution (Table 2). All PTSD patients were in format B, most controls of the original release in format C, and all somatoform patients in format A together with 20 controls. A rule that uses only the format separated the original PTSD and control groups with an AUC of 0.985. One format did not look like typical EEG: symmetric channels were uncorrelated and occipital alpha was absent. The supplementary batch of controls showed copied channels and a different high-frequency spectrum. These observations shaped every later decision.

Table 2. Export formats and groups (number of subjects).

Format	Amplitude step	PTSD	Controls	Somatoform
A	1 μ V	0	20	74
B	0.06 μ V	25	2, plus the supplementary batch and the 65+ group	0
C	about 0.01 μ V	0	45	0

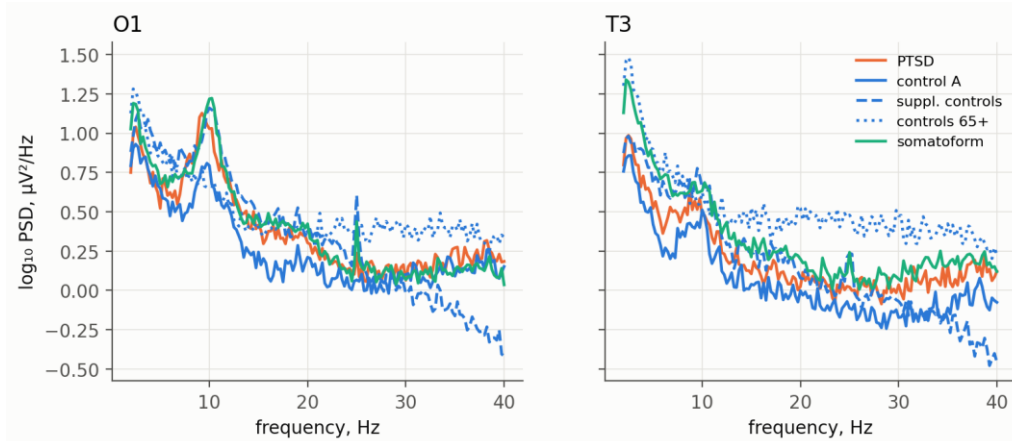


Figure 1. Median resting spectra at O1 and T3 by group. Older adults show a weaker and slower alpha peak, which is the main basis for separating them from patients. The supplementary controls fall off more steeply at high frequencies on every channel, a sign of a different recording setup rather than of physiology.

4.2 What the protocols taught us

The first protocol, using all channels and both rest and task, gave a high overall AUC that turned out to come almost entirely from the format C controls; against the format A controls it was at chance, and a model trained on metadata alone did equally well. Restricting the features to the most robust occipital and frontal measures removed the separation from comparable controls completely, an honest negative result. Behaviour in the Schulte tables was the most robust single signal, but older adults solve the

tables more slowly than patients, so behaviour mistakes them for PTSD. We kept behaviour out of the final model, since the task was EEG diagnostics and a third of the subjects had no behavioural data. The final protocols used resting EEG from all subjects, selected models by the official criterion and moved the threshold to the equal-error point. Table 3 summarises the sequence.

Table 3. Pre-registered protocols.

Protocol	Change	Outcome
1	rest and task features, all channels	high AUC explained by export format
2	rest only, no O2, spectral slope, new batch	separation within format relied on T3 and high frequencies
3	only occipital and frontal features, 4–20 Hz	no separation from comparable controls
4–5	Schulte behaviour, task EEG	behaviour robust but confounded by age; left out
6–7	resting EEG, official criterion, balanced calibration	sensitivity and specificity balanced
8	weights for specificity groups, equal-error threshold	better AUC and specificity; submitted

4.3 Out-of-sample performance

Table 4 and Figure 2 show the out-of-sample results of the submitted model. The AUC against all controls under 65 was 0.86. Without the format C controls it fell to 0.77, and against the format A controls alone it was weaker still, which shows how much of the headline number depends on one unusual group. At the 0.5 threshold the model detected 84% of patients. Specificity was high in older adults and moderate in somatoform patients. Among young controls it depended strongly on the batch: almost perfect for format C and close to one half for format A.

Table 4. Out-of-sample performance of the submitted model (95% confidence intervals).

Comparison	AUC	Rate at P = 0.5
PTSD vs controls under 65	0.86 [0.80; 0.92]	specificity 0.72 [0.64; 0.80]
PTSD vs controls without format C	0.77 [0.66; 0.86]	
PTSD vs controls aged 65+	0.89	specificity 0.87 [0.74; 1.00]
PTSD vs somatoform disorders	0.79	specificity 0.61 [0.50; 0.72]
PTSD patients		sensitivity 0.84 [0.69; 0.96]

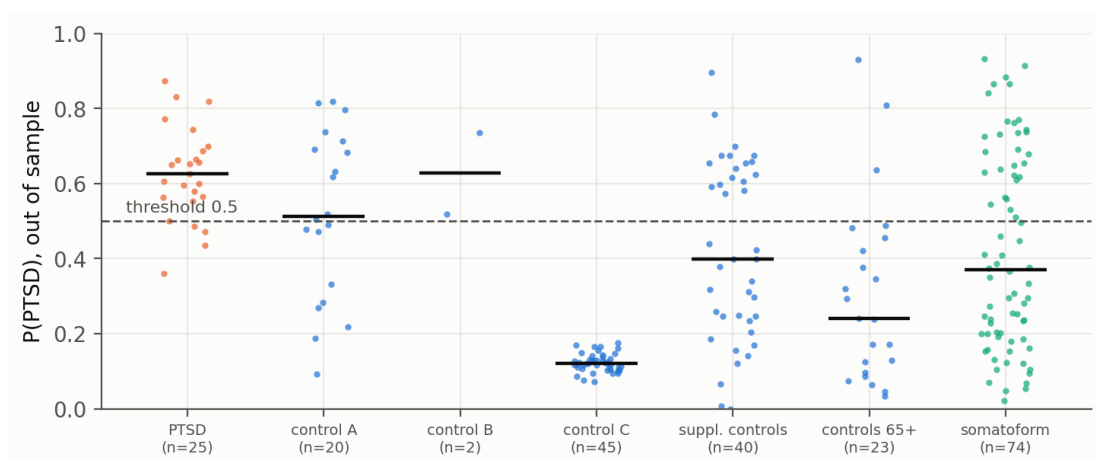


Figure 2. Out-of-sample probability of PTSD for every subject, by group. Each subject is predicted by a model that never saw them. Patients mostly lie above the 0.5 threshold and older adults below it. The format C controls form a tight cluster at low values, while control A and the supplementary batch spread across the threshold, which is where most false positives come from.

4.4 Confounders and robustness

Figure 3 compares the model with diagnostic models trained in the same folds. Export metadata alone and recording quality alone did not reproduce the separation from young controls of formats A and B.

Within a single export format the model still separated patients from controls, and in the permutation test none of the 60 runs with labels shuffled inside that format reached the observed AUC ($p = 0.016$). Within a batch, the features did not predict age, and the probability did not follow age or solve time.

Two results limit these conclusions. Without T3 and T4, the separation from young controls dropped close to chance, while the separation from older adults was unchanged. T3 lies next to the reference electrode, and its spectrum differs between recording batches, so for young controls physiology and batch cannot be told apart. When a control group was left out of the whole procedure and predicted as new data, the AUC against it fell by about 0.1; older adults remained the best separated group.

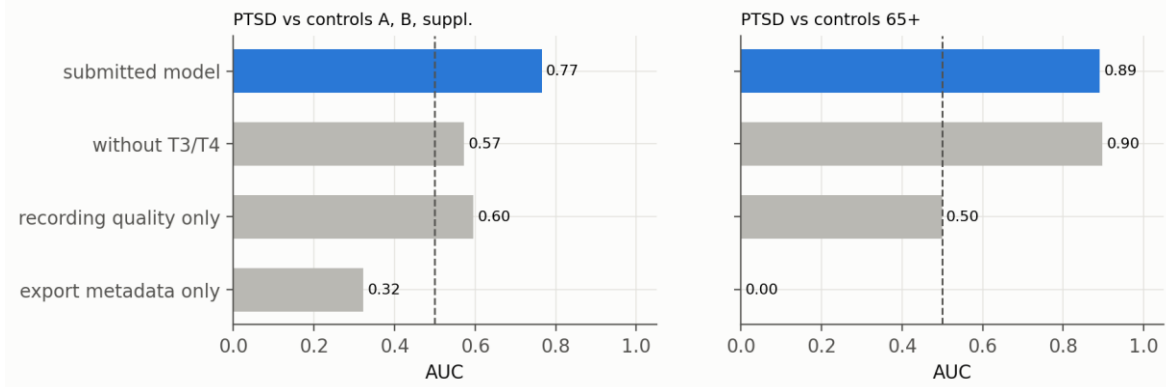


Figure 3. AUC of the submitted model and of diagnostic models evaluated in the same folds, against young controls of formats A and B (left) and against older adults (right). A model that only knows the export format or the recording quality does not reproduce the result. Removing T3 and T4 leaves separation from older adults intact but brings separation from young controls close to chance, which marks the temporal leads as the main remaining risk.

4.5 Features and physiology

Figure 4 shows how each feature differs between patients and young controls on each lead. Occipital alpha was not reduced in PTSD, unlike reports in the literature; in the model, alpha mainly separates older adults. The aperiodic spectrum was slightly flatter in patients, in line with the literature, but older adults show the same change. Frontal theta was slightly higher, which fits reduced vigilance and poor sleep. Frontal alpha asymmetry did not differ, possibly because Fp1 and Fp2 pick up eye movements, whereas the classic effect is described at F3 and F4.

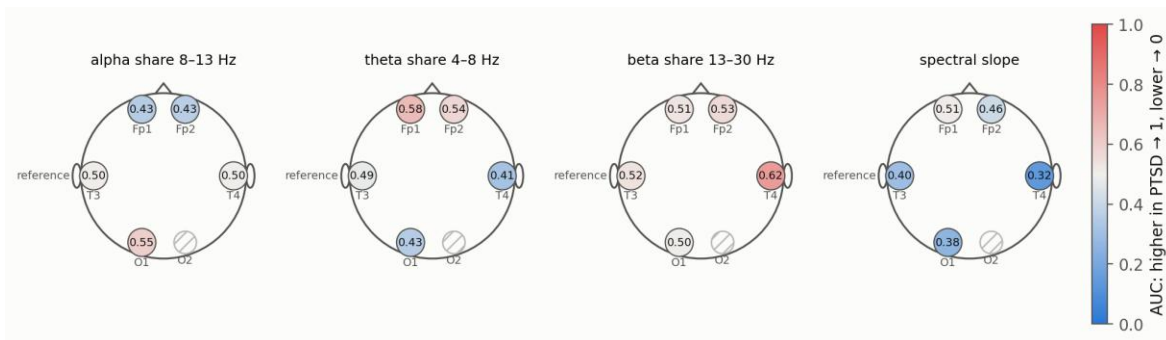


Figure 4. Difference between PTSD and young controls of formats A and B for each feature and lead, expressed as AUC (red: higher in PTSD, blue: lower; O2 is not used). The clearest effects are a flatter spectrum and more beta at T4. Both are shifted in the same direction in somatoform patients, so they describe a clinical profile shared by the two groups rather than a feature of PTSD.

To ask whether any feature is specific to PTSD, we compared patients with young controls, somatoform patients with controls of the same export format, and patients with somatoform patients. Only two features were reliably shifted in PTSD, a flatter spectrum and more beta at T4, and both were shifted in the same way in somatoform patients. Yet the model gave somatoform patients a lower probability than

healthy controls of the same format, so it does not separate them through this shared profile. Sex is a possible explanation, since all PTSD patients were men and the somatoform group included women.

4.6 Behaviour

Patients solved all five Schulte tables more slowly than controls, by roughly a quarter to a third, with the same warm-up and no slowing towards the end. On the 142 subjects with usable solve times, behaviour and EEG reached AUCs of 0.61 and 0.64 against all other subjects, and 0.75 together. The two signals complement each other: behaviour separates young controls, while EEG separates older adults, who are the slowest of all.

4.7 Auditory response to the metronome

A response to the metronome beats was present at rest ($p = 0.0005$), largest at Fp1 and Fp2, and absent at frequencies unrelated to the beat (Figure 5). It was not found during the Schulte task. On ERP CORE our loader reproduced the published mismatch negativity, and imitating the headset's 2 Hz hardware high-pass filter turned it into a three-phase wave at the frontal poles. In the hybrid simulation the method detected a deviant response only when it was clearly larger than a typical mismatch negativity. The confirmatory test missed its threshold ($p = 0.048$ against 0.025), so a response to the deviant tone is not confirmed.

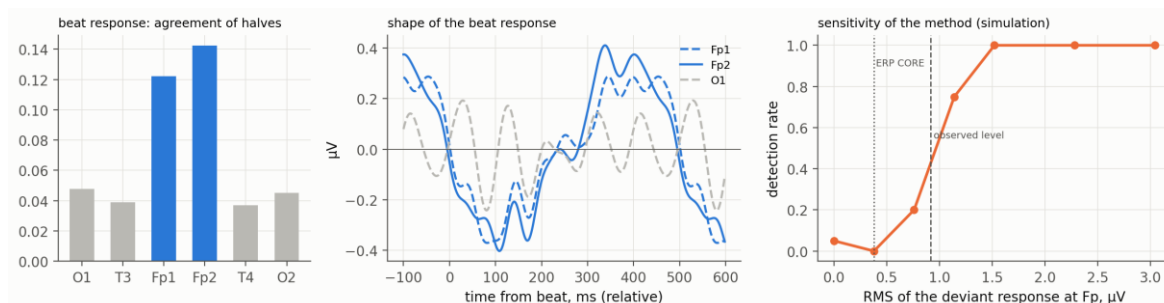


Figure 5. Response to the metronome recovered without stimulus markers. Left: agreement between the two halves of each recording, by channel; the frontal leads stand out. Centre: average response to a beat at Fp1 and Fp2, aligned on the half not used to find the timing. Right: detection rate of the deviant test in the hybrid simulation as a function of response size; responses of typical MMN size fall below what the method can detect.

5. Discussion

The first lesson of this dataset is that a high AUC can be obtained without looking at the brain. The export format alone nearly identified the patients, and the first model we built was largely detecting it. A useful evaluation therefore reports results for every combination of cohort, format and batch, includes models trained on metadata and quality as reference points, and fixes the analysis plan before seeing the result.

Within these constraints, one minute of dry EEG did carry information. Its most robust part was the separation from older adults, which rests on a well-known age effect: weaker and slower occipital alpha. This helps specificity in practice, but it is not a marker of trauma. Separation from young controls was weaker and depended on the temporal leads, where recording conditions and physiology are mixed. The features that did shift in PTSD were shared with somatoform patients, which fits a general profile of arousal or the effect of sedative medication such as benzodiazepines; medication was not recorded.

P should therefore be read as similarity to the EEG of this clinical cohort, not as the probability that a person has PTSD. The cohorts differ in more than the diagnosis: combat exposure, head injury, medication and sleep are all likely to differ between veterans and civilian volunteers, and each of them

changes resting EEG. With a PTSD prevalence of one in ten, about a quarter of positive results would be correct, so the model is not a diagnostic tool.

Limitations

The sample is small, especially the 25 patients, and all estimates come from the development data; the independent estimate is the organisers' closed test. Age was unknown for patients and sex for most subjects. The recording batches were not designed to be comparable, and with six channels and an ear reference the spatial resolution is limited. The metronome analysis could not use stimulus markers or central leads.

Recommendations

A trauma-specific marker would need a control group with combat exposure but without PTSD, one headset and export format for all groups, recorded medication, head injury, sleep, sex and age, symptom severity scales, longer and repeated recordings to measure reliability, and stimulus markers with central leads for event-related potentials.

6. Conclusion

We found that the data contain a shortcut strong enough to make almost any model look good, and built a simple classifier whose behaviour we could check against it. The model separates PTSD patients from healthy controls out of sample, separates them reliably from older adults, and keeps working within one export format. Its separation from young controls depends on the temporal leads and on the recording batch, and the shifts it relies on are shared with somatoform disorders. We hope that this kind of audit, rather than the headline AUC, is the useful result for future work on low-cost EEG in PTSD.

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Declarations

Ethics. The recordings were collected and pseudonymised by the hackathon organisers, who provided them for the competition. Information on ethical approval and informed consent is held by the organisers and is not reported here.

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Competing interests. The authors declare no competing interests.

Data and code availability

The EEG data belong to the hackathon organisers and are not public. Code, notebooks, saved validation results and the audit log are available at github.com/igor688-hub/neurodiagnosis-eeeg under the MIT license.

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Appendix A. Analysis settings

Table A1. Settings of preprocessing, features, model selection and evaluation.

Step	Setting
Resampling	125 Hz
Low-pass filter	40 Hz, zero-phase FIR
Windows	4 s with a 2 s step; Hann-tapered periodogram, 0.25 Hz resolution
Rejection per window and channel	SD below 0.5 μ V; peak-to-peak above 400 μ V; at least 3 samples at the amplitude limit; constant stretches of 0.1 s or longer with a 0.5 s margin; robust variance z-score above 3.5; copied channel
Minimum retained windows	5 per channel, otherwise the feature is missing
Spectrum	mean of log periodograms with the bias correction $\gamma / \ln 10$
Bands	theta 4–8 Hz, alpha 8–13 Hz, beta 13–30 Hz; reference range 4–30 Hz
Aperiodic fit	3–30 Hz excluding 7–14 Hz
Alpha peak	searched in 7–13 Hz; at least 0.3 $\log_{10}$ units above the background; frequency as the centroid within ± 1 Hz
Missing values and scaling	training median; training mean and standard deviation
Candidate grid	4 feature sets (with or without T3/T4, with or without slope) $\times C \in \{0.01, 0.03, 0.1, 0.3, 1\} \times$ negatives (controls, or controls and somatoform) $\times$ weight of specificity groups $\in \{1, 3\}$
Inner cross-validation	5-fold, stratified by group and stratum
Outer validation	leave-one-group-out over 221 independent groups
Calibration	Platt scaling with balanced class weights, shifted to the equal-error point
Uncertainty	2000 bootstrap resamples of groups, percentile 95% intervals
Permutation test	60 full nested runs with labels shuffled within export format B
Random seed	42